

Model review and revisions

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I reviewed the model and calibration, including preferences, credit, demographic accounting, and the empirical targets.

- Returned to a Stone–Geary housing requirement that applies only to parents, retaining nonlinear benefits from children.
- Tested equivalence scales, changing housing shares, and the utility normalization.
- Rebuilt the first-birth housing evidence and national input measures.
- Revised adult entry and tested how purchase timing affects the calibration.

Household decisions

Overlapping generations of households differ in age, earnings, wealth, housing tenure, and family size.

- During fertile ages, households choose whether to attempt a birth; success becomes less likely with age.
- They choose consumption, saving, housing size, and whether to rent or own; renters pay rent, while owners make a down payment and face selling costs.
- Persistent earnings risk and borrowing limits shape saving; a first-birth fixed cost and taste shocks shape fertility.
- Households retire, face mortality risk, and value bequests.

Children and housing needs

The first child requires basic space at home; each additional child raises the goods needs of a larger family.

$$u(c, s; m) = \frac{\left[\frac{c^{\alpha_0} s^{1-\alpha_0}}{e(m)} \right]^{1-\sigma}}{1-\sigma} + \psi m^{1-\gamma},$$
$$e(m) = \left(\frac{2 + 0.7m}{2} \right)^{0.7}, \quad s = \eta_\tau [h - h_P \mathbf{1}\{m > 0\}],$$
$$\eta_R = 1, \quad \eta_O = \chi.$$

Here h is rooms, m is children at home, and $\tau \in \{R, O\}$ denotes tenure. Parents require $h > h_P$; χ raises housing services for owners. The equivalence scale $e(m)$ increases family consumption needs. Child benefits are zero at $m = 0$; ψ sets their level and γ controls diminishing marginal benefits.

Demographic and market equilibrium

Children leaving home and new adult households entering the economy are separate transitions.

- Children leave dependency independently; adult entry follows earlier births with a lag.
- Positive estates finance entrant assets; any remainder leaves the economy.
- Stationary calibration normalizes population to one: the house price satisfies birth renewal, and housing clearing determines H_0 .
- A balanced payroll tax finances equal retiree pensions.

Housing around the first birth

We found a mistake in the earlier regression and rebuilt the PSID event study using interview dates and household status before birth.



At years 3–4, rooms rise by **1.465** (SE 0.050), ownership by **26.886 pp**, and recent moving falls by **21.147 pp**.

Reference person or spouse before birth. Baseline: years $-3/-2$; person and year effects; survey weights; 95% pointwise intervals.

Credit timing

With $\phi = 0.8$, a purchase requires a 20% down payment. Purchase timing determines which resources count toward it. The soft rule includes current-period income in the purchase screen; the other rules use:

$$\text{Hard: } A \geq (1 - \phi)Q,$$

$$\text{Quarter-saving: } A + \frac{1}{4} [b' - (A - Q)] \geq (1 - \phi)Q.$$

Here A is financial wealth plus net proceeds from selling the previous home, Q is the cost of the chosen home, and b' is the end-of-period financial balance. Net saving is $b' - (A - Q)$.

For a home costing 100 with 10 available at closing, the hard rule blocks the purchase. The quarter-saving rule meets the 20-unit equity requirement if net saving is 40:

$$10 + 0.25 \times 40 = 20.$$

This approximates saving before purchase within a four-year period. Under quarter-saving, income helps only if saved. Hard and quarter-saving also impose an end-of-period net-estate floor for buyers; this is an additional change from soft.

More credit raises ownership but not births

Quarter-saving rule; financed share 80% vs. 100%; house price and parameters held fixed

	80% financing	100% financing
Ownership, ages 30–55 (%)	64.6	74.0
First-birth flow (relative change)		–0.5%
Completed fertility	2.100	2.090
<i>Households entering at ages 18–21 (same households in both)</i>		
Ownership (%)	10.5	31.4
First-birth probability (%)	25.5	25.8

Recalibrating under the hard and quarter-saving rules gives the same sign: first births fall by about 0.5%.

Why the response is small: what we have checked

- Rent per room equals the owner's cost per room, and χ does not depend on children. Credit makes buying easier but does not make a child's space cheaper.
- 60% of renting parents are at the six-room cap. Without the cap the median would rent 7.3 rooms. Holding saving fixed, the cap costs them about 2% of consumption in that period.
- The parent requirement h_P is the same for one, two or three children.

These are candidate explanations. We have not yet shown how much of the small response each one accounts for.

Selling costs and small owner homes

Exploratory runs, not recalibrated. Change in completed fertility from 80% to 100% financing

Selling cost	6%	4%	2%	0%
Change in completed fertility	-0.010	-0.009	-0.005	+0.012

Owner sizes (rooms)	Change in completed fertility	Fit loss at 80%
2-10 (current)	-0.010	51.6
4-10	-0.007	57.5
6-10	+0.001	284.3

- At 100% financing more young childless households buy two-room homes, which are too small for parents.
- The sign turns positive only without selling costs. Dropping small homes removes the negative sign and worsens the fit.

The selling-cost runs also add one required room per additional child.

Parameters and target moments

2007 stationary economy: ten parameters are estimated jointly from ten moments.

Parameter	Economic interpretation	Target moments
β	Patience	Wealth / earnings (6.927)
θ_0	Bequest weight	Annual bequests / wealth (0.729%)*
χ	Owner housing premium	Ownership, ages 30–55 (67.626%); mean rooms (5.729)*
κ_T	Tenure tastes	Ownership, ages 30–55; recent-parent gap (12.761 pp)
h_P	Basic space for parents	First-birth rooms (1.465)*; mean rooms
ξ	First-birth cost	Childlessness, ages 40–44 (19.828%); children at 25 (0.810)
κ_1	First-birth taste dispersion	Mean first-birth age (25.976); children at 25
κ_C	Later-birth taste dispersion	One child among mothers (21.366%); children at 25
ψ	Child-benefit level	Childlessness; children at 25
γ	Child-benefit curvature	One child among mothers (21.366%); children at 25

*Recomputed: mean rooms (AHS), bequests (SCF), and first-birth rooms (PSID). Also updated pension/earnings, depreciation and property tax. Parameters jointly affect the moments; H_0 follows from housing clearing and θ_1 is fixed externally.

Stone–Geary calibration

2007 stationary economy; 80% mortgage financing

Moment	Target	Soft*	Hard	Quarter-saving
Completed fertility	2.100	2.100	2.100	2.100
Childlessness, ages 40–44 (%)	19.828	20.222	20.413	20.546
One child among mothers, ages 40–44 (%)	21.366	21.264	20.907	20.844
Mean first-birth age (years)	25.976	26.032	26.007	26.057
Wealth / earnings	6.927	6.551	6.610	6.645
Annual child bequests / wealth (%)	0.729	0.685	0.684	0.685
Mean occupied rooms	5.729	5.957	6.241	6.131
Ownership, ages 30–55 (%)	67.626	66.527	60.878	64.572
First-birth rooms difference	1.465	1.324	1.063	1.167
Recent-parent ownership difference (pp)	12.761	11.825	10.125	11.683
Children at age 25 (capped at three)	0.810	0.528	0.524	0.521
Validation moments	Target	Soft*	Hard	Quarter-saving
First births at age 30+ (%)	24.928	23.448	23.078	23.395
Older wealth/income p90/p50	3.516	2.968	2.861	2.937
Family rooms difference	0.385	0.304	0.321	0.316

Soft*: best fit before the search stopped; final numerical check pending.

Parameter estimates

Estimates under the three purchase rules; H_0 follows from housing clearing.

Parameter	Economic interpretation	Soft*	Hard	Quarter-saving
β	Annual discount factor	0.967	0.967	0.968
χ	Owner housing-service premium	1.097	1.068	1.091
ξ	First-birth fixed utility cost	0.352	0.385	0.397
κ_1	First-birth taste dispersion	0.124	0.123	0.128
κ_C	Later-birth taste dispersion	0.363	0.350	0.348
θ_0	Bequest weight	0.106	0.106	0.109
γ	Child-benefit curvature	0.101	0.101	0.098
κ_T	Tenure-choice dispersion	0.014	0.014	0.015
ψ	Child-benefit level	0.178	0.172	0.174
h_P	Basic space requirement for parents (rooms)	2.504	2.562	2.492
H_0	Housing supply coefficient (derived)	6.757	7.546	7.289

H_0 is derived at normalized population $N_0 = 1$. Fixed inputs: $\sigma = 2$; $\alpha_0 = 0.733$; annual interest 2%; financed share $\phi = 0.8$.

Calibration comparison

- The soft rule fits best. Targets and weights are the same across specifications.
- Mean rooms: 5.957 (soft), 6.241 (hard), 6.131 (quarter-saving), against a target of 5.729.
- First-birth rooms difference: 1.324, 1.063 and 1.167, respectively, against 1.465.
- Children at age 25: 0.528, 0.524 and 0.521, against 0.810. Early fertility remains low in all three.
- Each rule is recalibrated. Hard and quarter-saving also require nonnegative net wealth after purchase.